

TrES-1b

R-1076

Benchmark run · workflow W8-benchmark-deep · record deep-bench_TRES-1_140530 · created 2026-07-30T14:47:08+00:00

BENCHMARK: ACCURATE — fit consistent with published values

Results

Mid-Transit Time (Tmid)	2456807.7387 +/- 0.0012 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.128 +/- 0.012
Transit depth (Rp/Rs)^2	1.64 +/- 0.32 [%]
Orbital Inclination (inc)	89.88 +/- 0.71
Airmass coefficient 1 (a1)	1.189 +/- 0.016
Airmass coefficient 2 (a2)	-0.1423 +/- 0.0026
Scatter in the residuals of the lightcurve fit is	1.26 %
Best Comparison Star	None
Optimal Aperture	4.5
Optimal Annulus	11.29
Transit Duration (day)	0.1028 +/- 0.0015

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.128 ± 0.012 vs 0.1358 (deviation 0.65σ)
Duration fit vs published	0.1028 d vs 0.1042 d (deviation 0.91σ)
Mid-time O–C	-3.1 min
Depth SNR	10.7
Residual scatter	1.26 %

Light curve

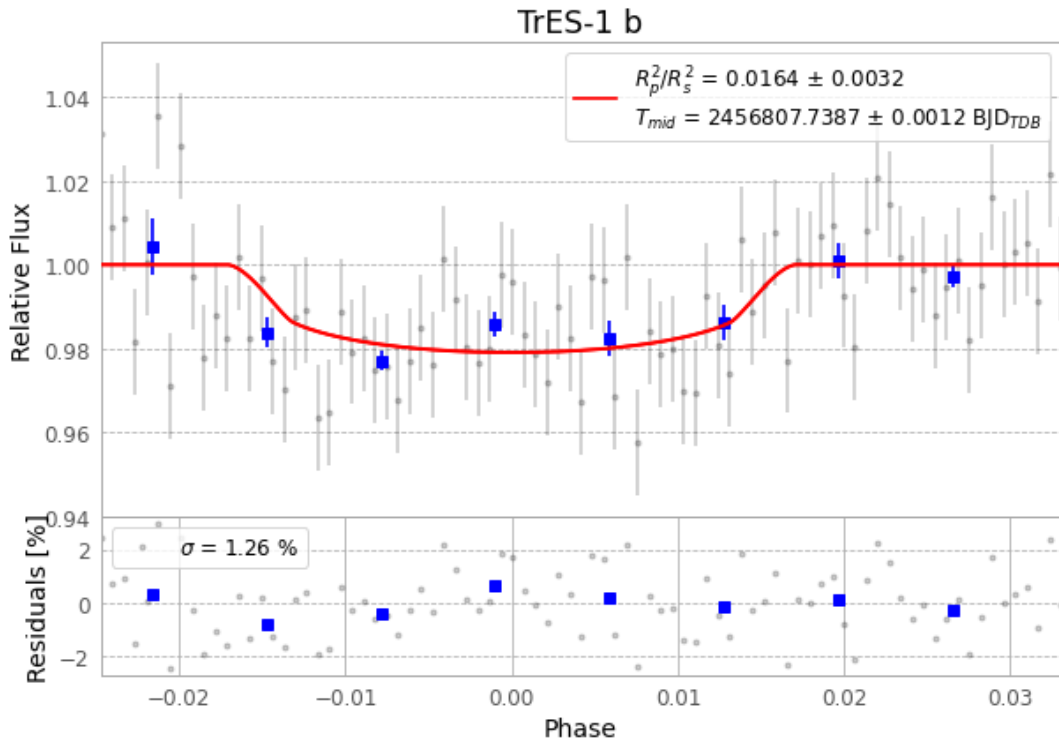


Figure 1 — FinalLightCurve_TrES-1 b_2014-05-29.png (SHA-256 5485fe1836f53fd7...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [135, 196], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 051d855b5bf099f4...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ba3ef8b

Data provenance

85 FITS frames (56 MB), TRES-1140530035112.FITS → TRES-1140530080313.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-TRES-1-140530.log (b4439a72ecdf...), FinalLightCurve_TrES-1 b_2014-05-29.png (5485fe1836f5...), FinalLightCurve_TrES-1 b_2014-05-29.csv (d62b233f9544...)

Compute resources

Wall time 140.2 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-30 14:47Z.